

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE

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Subject Code: BTCOC603_Y23 Subject Name: Machine Learning
Created By: @prantehare Total Marks: 60 Marks

Study & Revision Guide: Question No. 1 is compulsory (12 marks). Answer any 4 questions from Questions 2 to 6. This manual provides detailed explanations, comparative grids, and mathematically rigorous answers mapped with structural core patterns.

QUESTION NO. 1 (Compulsory Objective Pack) [12 Marks]

No.	Question Statement & Options	Correct Answer & Technical Context
1	A model is trained to predict house prices from features like size, number of rooms, and location. The training data includes both the features and the actual sale prices. Which type of machine learning problem is this? a. Regression b. Classification c. Supervised Learning d. Reinforcement Learning	c. Supervised Learning It involves labeled datasets containing both target outputs (sale prices) and input attributes, making it a supervised system. (Note: Option 'a' is a sub-type, but 'Supervised Learning' broadly covers the learning framework).
2	You train a classifier and observe that it performs very well on the training data but poorly on unseen test data. Which situation does this most likely indicate? a. Underfitting b. High Bias & Low Variance	d. Overfitting Overfitting occurs when a complex hypothesis maps structural training noise perfectly but completely fails to generalize on clean, unseen evaluation splits.

No.	Question Statement & Options	Correct Answer & Technical Context
	c. Good Generalisation d. Overfitting	
3	Which of the following is the main goal of using a validation set during model development? a. To Provide Completely Unseen data for Final Reporting b. To Train the Model Parameters c. To Tune Hyperparameters & Select the Best Models d. To Increase the Size of the Training Set	c. To Tune Hyperparameters & Select the Best Models The validation set is used to optimize structural hyper-configurations (such as depth, tree limits, or regularizer scales) without biasing target evaluation tracks.
4	In a binary classification problem, which metric is most appropriate when the positive class is rare, and the cost of missing a positive case is very high? a. Precision b. Accuracy c. Recall d. AUC ROC	c. Recall Recall measures the system's ability to catch all actual positive instances, making it critical for high-stakes scenarios like medical screening or fault detection.
5	Which of the following best describes the bias-variance trade-off in machine learning models? a. Bias is Always Harmful, Variance is Always Beneficial b. Bias & Variance are Unrelated Concepts c. Reducing Bias often increases variance and vice versa d. Increasing Model Complexity always reduces both Bias & Variance	c. Reducing Bias often increases variance and vice versa As a model's capacity scales up, it fits training contours closer (reducing bias), but becomes highly sensitive to dataset fluctuations (increasing variance).
6	What is the primary purpose of regularisation techniques such as L1 or L2 regularization in machine learning models?	a. To Prevent Overfitting by Controlling Model Complexity Regularization penalizes overly large weight coefficients, keeping the learned decision boundary smooth and helping prevent overfitting.

No.	Question Statement & Options	Correct Answer & Technical Context
	a. To Prevent Overfitting by Controlling Model Complexity b. To Reduce The Size of the Dataset c. To Guarantee Perfect Accuracy on the Training Dataset d. To Decrease Model Training Time to Zero	
7	Which algorithm is commonly used for both classification and regression problems? a. k-Means b. Decision Tree c. PCA d. Apriori	b. Decision Tree Decision Trees branch out split selections using either categorical impurity scores (Gini/Information Gain) or continuous variance-reduction tests.
8	What is the role of an activation function in a neural network? a. Reduce computational cost b. Introduce non-linearity c. Initialize weights d. Normalize input data	b. Introduce non-linearity Without non-linear transformations (like ReLU or Sigmoid), a deep neural network collapses mathematically into a single basic linear model.
9	Which algorithm uses "Gradient Descent" for optimisation? a. K-Nearest Neighbours b. Linear Regression c. Naive Bayes d. Random Forest	b. Linear Regression Linear Regression models can iteratively minimize their Mean Squared Error (MSE) loss surface using standard gradient descent updates.
10	In reinforcement learning, what does the agent learn from? a. Labelled data b. Unlabelled data c. Rewards and penalties d. Feature correlations	c. Rewards and penalties Reinforcement agents learn by interacting with an environment, using trial-and-error signals to discover a policy that maximizes cumulative reward over time.
11	Which method is used for handling missing data? a. Deleting all incomplete rows b. Replacing with random values	c. Imputation (e.g., mean, median) Imputation updates missing records with statistical indicators like the feature mean, median, or values predicted by a tracking model.

No.	Question Statement & Options	Correct Answer & Technical Context
	c. Imputation (e.g., mean, median) d. Ignoring the issue	
12	Which evaluation metric is suitable for imbalanced classification? a. Accuracy b. AUC-ROC c. Mean Squared Error d. R-squared	b. AUC-ROC AUC-ROC measures performance across all classification thresholds, tracking true positive and false positive rates even when class distributions are highly skewed.

QUESTION NO. 2 (Generalization & Optimization Basics)

[12 Marks]

A) Compare the bias-variance trade-off. How do they affect model performance? [6 Marks]

The bias-variance trade-off describes the delicate balance between a model's complexity and its ability to generalize effectively to new data.

Attribute	Bias Error	Variance Error
Definition	Error caused by overly simplistic assumptions in the learning algorithm.	Error caused by extreme sensitivity to small fluctuations in the training dataset.
Model State	Leads to Underfitting ; the model cannot capture the true underlying data patterns.	Leads to Overfitting ; the model captures noise and random variations as if they were true patterns.
Error Profile	Shows high error on both training data and testing datasets.	Shows low error on training data but high error on unseen evaluation sets.
Complexity Path	Occurs when model complexity is too low (e.g., fitting a straight line to highly curved data).	Occurs when model complexity is too high (e.g., using a high-degree polynomial for simple linear data).

Impact on Total Performance: The total expected error of a model is the sum of **Bias² + Variance + Irreducible Noise**. Minimizing total error requires finding an optimal balance point: increasing complexity reduces bias but increases variance, while lowering complexity does the reverse.

B) Explain how gradient descent optimises a model.**[6 Marks]**

Gradient Descent is an iterative optimization algorithm used to find the minimum of a cost function (such as Mean Squared Error) by updating a model's parameter weights.

Step-by-Step Working Principle:

1. **Initialization:** The weights (w) and biases (b) are initialized to small random values or zeros.
2. **Compute the Loss:** The model runs predictions across data points and calculates the objective loss value via the cost function $J(w)$.
3. **Calculate Gradients:** The algorithm computes the partial derivatives of the cost function with respect to each parameter weight. This gradient value represents the slope of the steepest ascent on the loss surface:

$$\nabla J(w) = \partial J / \partial w$$

4. **Update Parameters:** The weights are updated by moving a small step in the opposite direction of the gradient (downward toward the minimum). This step size is controlled by the learning rate parameter (η):

$$w := w - \eta \times \nabla J(w)$$

5. **Convergence:** This loop repeats until the gradients approach zero or the loss stops changing significantly, indicating the model has successfully reached a local or global minimum.

QUESTION NO. 3 (Ensembles & Regularization)**[12 Marks]****A) Describe the working of Random Forest. Why is it called "forest"?****[6 Marks]**

Random Forest is a popular ensemble learning method that builds multiple individual decision trees to perform classification or regression tasks.

How it Works:

- **Bootstrapping (Bagging):** The algorithm creates multiple subsets of the original training data by sampling rows randomly with replacement. Each tree is trained on a different subset.
- **Random Feature Selection:** When splitting a node inside a tree, the algorithm considers only a random subset of the available features rather than checking all of them. This decorrelates the trees from one another.

- **Independent Growth:** Every decision tree is grown to its maximum depth without pruning, allowing individual trees to have high variance but low bias.
- **Ensemble Voting/Aggregation:** To make a final prediction on new data, the model combines the outputs of all its trees: it uses a majority vote for classification tasks and averages the predictions for regression tasks.

Why it is called a "Forest": It earned the name "forest" because it doesn't rely on a single decision maker; instead, it aggregates a large collection (an ensemble) of independent decision trees working together to produce a more accurate and stable final prediction.

B) What is regularisation? Explain L1 (Lasso) vs. L2 (Ridge).

[6 Marks]

Regularization refers to a set of techniques used to prevent overfitting by adding a penalty term to the cost function, discouraging models from learning overly complex patterns.

L1 Regularization (Lasso - Least Absolute Shrinkage and Selection Operator):

- Adds a penalty proportional to the sum of the absolute values of the weights: **Penalty = $\lambda \sum |w_j|$** .
- Forces less important feature weights to become exactly zero, effectively performing automatic feature selection and producing sparse models.

L2 Regularization (Ridge Regression):

- Adds a penalty proportional to the sum of the squared values of the weights: **Penalty = $\lambda \sum (w_j)^2$** .
- Shrinks all weight coefficients close to zero but never makes them exactly zero, preserving all features while reducing overall model sensitivity.

QUESTION NO. 4 (Dimensionality, SVM & Deep Networks) [Solve Any Two • 12 Marks]

A) Explain principal component analysis (PCA) and its applications.

[6 Marks]

Principal Component Analysis (PCA) is an unsupervised linear dimensionality reduction technique used to project high-dimensional data into a lower-dimensional space while preserving as much variance as possible.

Core Mathematical Steps: First, the data is standardized so it has a mean of zero and a variance of one. The algorithm computes the covariance matrix to look for correlations between features. It then calculates the *eigenvectors* and *eigenvalues* of this matrix. The eigenvectors represent the new

orthogonal axes (called Principal Components) ordered by how much variance they explain, allowing the lower-ranked components to be safely dropped.

Primary Applications:

- **Data Visualization:** Projects complex, high-dimensional datasets down to 2D or 3D spaces so they can be plotted and analyzed visually.
- **Feature Reduction & Compression:** Compresses large datasets to reduce storage needs and speed up the training time of downstream models.
- **Noise Filtering:** Drops lower-ranked principal components that primarily capture random noise, helping improve model performance.

B) What are support vectors in SVM? How does kernel trick work?

[6 Marks]

Support Vectors: In a Support Vector Machine (SVM), support vectors are the data points from the training set that lie closest to the decision boundary (the separating hyperplane). They are the critical points that define the position and orientation of the boundary; removing or changing any other data points has no effect on the margin.

The Kernel Trick: When data is linearly inseparable in its original lower-dimensional space, the kernel trick maps the data into a higher-dimensional feature space where a clean linear boundary becomes possible. Instead of manually transforming every coordinate—which can be slow and computationally expensive—the algorithm uses a mathematical *kernel function* (like RBF or Polynomial) to calculate the inner products between points directly in the high-dimensional space. This allows the model to learn complex, non-linear boundaries efficiently without ever explicitly computing the high-dimensional coordinates.

C) Explain backpropagation in neural networks with the chain rule.

[6 Marks]

Backpropagation is the fundamental algorithm used to train neural networks by calculating the gradient of a loss function with respect to all the network's weights and biases.

Mechanism & Use of the Chain Rule: Training starts with a forward pass, where inputs flow through the network layers to generate a prediction, which is then compared against the true target value to calculate an overall error loss (E). During the backward pass, the algorithm computes how much each individual weight contributed to that error by working backward from the output layer to the input layer.

Because weights are nested deep within consecutive layers of activation functions, backpropagation relies on the mathematical **Chain Rule** of calculus to compute these derivatives. For example, to find how the error changes with respect to a weight w in an early layer, it multiplies the sequential derivatives along that path:

$$\partial E / \partial w = (\partial E / \partial y) \times (\partial y / \partial z) \times (\partial z / \partial w)$$

These calculated gradients are then passed to an optimizer like SGD or Adam to update the weights and systematically minimize the model's error over time.

QUESTION NO. 5 (Validation, Clustering & Ensembles)

[Solve Any Two • 12 Marks]

A) Explain ensemble learning with examples.

[6 Marks]

Ensemble learning is a machine learning strategy that combines the predictions of several base models to build a single, more robust and accurate model.

Primary Strategies with Examples:

- **Bagging (Bootstrap Aggregating):** Trains multiple base models independently in parallel on different random subsets of the data, then averages their predictions to reduce overall variance. *Example: Random Forest.*
- **Boosting:** Trains base models sequentially, where each new model focuses on correcting the errors made by the previous ones, helping reduce bias. *Examples: AdaBoost, Gradient Boosting, XGBoost.*
- **Stacking:** Trains several different types of models (like a linear regression and a decision tree) in parallel, then feeds their predictions into a final meta-model that learns how to best combine them.

B) What is the purpose of training, validation, and test datasets?

[6 Marks]

To build reliable machine learning models, datasets are typically split into three distinct subsets, each serving a specific role:

1. **Training Dataset:** The primary dataset used by the learning algorithm to fit internal parameters, such as the weights and biases in a neural network or the splits in a decision tree.
2. **Validation Dataset:** Used during development to evaluate the model's performance while tuning hyperparameters (like choosing the optimal learning rate or tree depth). It acts as a guide to prevent overfitting before final deployment.
3. **Test Dataset:** A completely independent partition used only after training and hyperparameter tuning are fully complete. It provides an unbiased evaluation of how well the final model generalizes to real-world, unseen data.

C) How does K-Means clustering work?**[6 Marks]**

K-Means is an unsupervised iterative clustering algorithm used to partition a dataset into **K** distinct, non-overlapping groups based on feature similarity.

Step-by-Step Algorithm:

1. **Initialization:** The user defines the number of clusters (**K**), and the algorithm randomly selects **K** data points to act as the initial cluster centers (centroids).
2. **Assignment Step:** The algorithm calculates the distance (typically using Euclidean distance) from every data point to all **K** centroids, assigning each point to its nearest center.
3. **Update Step:** For each of the **K** groups, the algorithm recalculates the centroid position by taking the mean average of all the coordinates of the data points assigned to that cluster.
4. **Convergence Loop:** The assignment and update steps repeat in a loop until the centroid positions stop changing or the assignments stabilize, indicating the clusters are fully formed.

QUESTION NO. 6 (Regression, Sampling & Optimization)**[Solve Any Two • 12 Marks]****A) Explain how linear regression works, including the cost function and optimisation method.****[6 Marks]**

Linear Regression models the relationship between dependent target variables and independent predictor features by fitting a straight linear equation to the observed data points.

The Core Model: The prediction equation is formulated as:

$$y_{\text{pred}} = w_1 \times x_1 + w_2 \times x_2 + \dots + b$$

The Cost Function: To measure how well the model fits the data, it uses the **Mean Squared Error (MSE)** cost function, which calculates the average value of the squared differences between the true targets (**y**) and the model's predictions (**y_pred**):

$$J(w, b) = (1 / 2n) \times \sum (y_{\text{pred}} - y)^2$$

The Optimization Method: The model optimizes its parameters using **Gradient Descent**. It computes the partial derivatives of the MSE cost function with respect to the weights and biases, then iteratively updates those values in the opposite direction of the gradient step by step until it finds the optimal line that minimizes overall error.

B) Explain SMOTE for handling imbalanced datasets.**[6 Marks]**

****SMOTE (Synthetic Minority Over-sampling Technique)**** is a preprocessing algorithm used to resolve class imbalance issues by generating synthetic examples of the minority class, rather than simply duplicating existing records.

How the Algorithm Works:

- The algorithm analyzes minority class data points and identifies their nearest neighbors using a distance metric (typically **K-Nearest Neighbors**).
- It randomly selects one of these neighbors for a given minority point.
- To create a new synthetic record, it draws a line segment connecting the minority point and its chosen neighbor, then picks a random point along that line to serve as the new data instance.

By filling in the feature space between existing examples rather than making exact copies, SMOTE creates smoother, more robust decision boundaries and helps prevent models from overfitting to the minority class.

C) What is gradient descent? Differentiate batch, stochastic, and mini-batch variants.**[6 Marks]**

Gradient descent is an iterative optimization algorithm used to minimize a model's cost function by updating its weights based on calculated derivatives. It features three primary variants that differ in how much data they process per update:

1. **Batch Gradient Descent:** Calculates gradients across the entire training dataset before performing a single parameter update. While this provides a smooth, stable path to convergence, it can become slow and memory-intensive for large datasets.
2. **Stochastic Gradient Descent (SGD):** Calculates gradients and updates parameters using just a single randomly chosen data point at a time. This approach is fast and memory-efficient, but its updates fluctuate heavily, causing the loss to jump around rather than settle smoothly.
3. **Mini-Batch Gradient Descent:** Strikes a balance by splitting the dataset into small, manageable groups (called mini-batches, typically sized between 32 and 256 points). It updates model weights after processing each mini-batch, combining the stable convergence of Batch descent with the speed and efficiency of SGD.